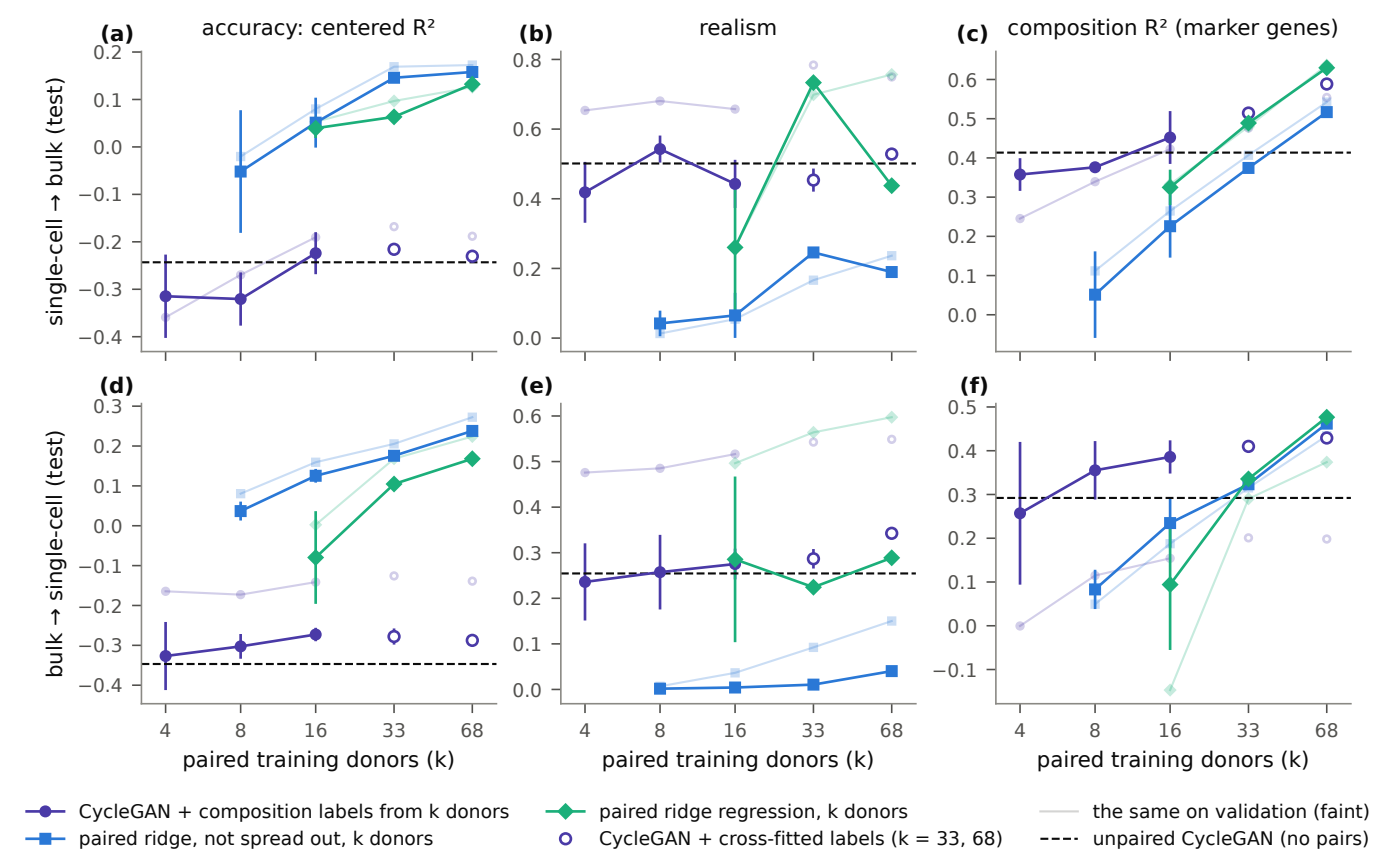
(a)
accuracy: centered R²
(b)
realism
(c)
composition R² (marker genes)
single-cell → bulk (test)
(d)
(e)
(f)
bulk → single-cell (test)
4
8
16
33
68
paired training donors (k)
CycleGAN + composition labels from k donors
paired ridge, not spread out, k donors
paired ridge regression, k donors
CycleGAN + cross-fitted labels (k = 33, 68)
the same on validation (faint)
unpaired CycleGAN (no pairs)